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Difference between DBMS and RDBMS:

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DBMS (Database Management System) is a basic way to store data in individual files with no connection between them, making data retrieval tedious and time-consuming¹.

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RDBMS (Relational Database Management System) has relationships between data¹. It allows creating relationships between different pieces of data, enabling faster data retrieval, especially with large datasets¹. RDBMS reduces redundancy, supports multiple users with high security, and is optimized for larger amounts of data². Examples include MySQL¹.

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Primary Key and Foreign Key:

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A **Primary Key** uniquely identifies rows in a table². It ensures each row is distinct and cannot be duplicated, and it does not allow any empty or null values². Think of a role number in a class².

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A **Foreign Key** helps connect data across tables². It points back to the primary key in another table, creating a link between the two tables². For example, a role number in a course table could link back to the primary key (role number) in a student table².

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Constraints and their types: Constraints are like rules set up for data in tables to help keep data accurate and reliable².

-

NOT NULL: A column cannot be empty²....

-

UNIQUE: Ensures no two values in a column are the same³.

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PRIMARY KEY: Combines NOT NULL and UNIQUE constraints, meaning there should be only one field that uniquely identifies records³.

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FOREIGN KEY: Links tables together³.

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CHECK: Allows setting a condition for the data in a column, like accepting only ages over 18 in a voting age column³.

-

DEFAULT: Plays an important role when you want a record to take a default value if none is specified³.

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DDL and DML commands:

-

DDL (Data Definition Language) commands define the structure of the database by creating, altering, or deleting tables and other objects³. They do not deal with the actual data but with the database structure³.... Examples include `CREATE` (make new tables), `ALTER` (modify tables), and `DROP` (delete a whole table)³.

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DML (Data Manipulation Language) commands work with the actual data once the table structure is defined⁴. They are used to insert, update, or delete data⁴. Examples include `INSERT` (add a new row), `UPDATE` (change data), and `DELETE` (remove specific data)⁴.

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Difference between DELETE, DROP, and TRUNCATE:

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DELETE: Removes specific rows from a table based on certain conditions⁴. It allows you to roll back the deletion⁴.

-

TRUNCATE: Quickly clears out all rows from a table, leaving the table structure intact⁴. It is fast and does not allow rolling back⁴. Use it when you don't need any data but want to keep the structure⁴.

-

DROP: Takes down the entire table structure, including all the data⁵. It is permanent and cannot be undone⁵. Use it only if you know you won't need the table or data again⁵.

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Difference between GROUP BY and ORDER BY:

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GROUP BY: Used to group data into buckets based on one or more columns⁵. It is typically used with aggregate functions (like `SUM`, `AVG`, `COUNT`) to calculate summaries for each group⁵.

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ORDER BY: Used for sorting rows in a particular order based on one or more columns⁵. You can sort in ascending or descending order⁵.

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Difference between WHERE and HAVING Clause:

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WHERE Clause: Filters individual rows based on a specific condition **before** any grouping occurs⁶.

-

HAVING Clause: Filters groups created by the `GROUP BY` clause based on aggregate results **after** grouping occurs⁶.

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Aggregate functions: These are tools that help perform calculations on data in a database⁶.

-

COUNT: Counts the number of rows or non-null values in a column⁶....

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SUM: Adds up all values in a numeric column⁷.

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AVG: Calculates the average of a numeric column⁷.

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MIN: Finds the minimum value in a numeric column⁷.

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MAX: Finds the maximum value in a numeric column⁷.

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Indexing and Clustered Index:

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Indexing is like organizing data in a specific manner to make data retrieval fast and efficient⁷. Without an index, searching for specific data is slow⁷.

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A **Clustered Index** is like physically organizing the data rows in the table in the order of the index⁷. Searching using a clustered index allows the database to quickly jump to the relevant data range⁷.

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Normalization and Normal Forms:

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Normalization in databases works by organizing data efficiently, minimizing redundancy (duplicate information), and preventing issues when inserting, deleting, or updating records⁷.... It reduces duplication and prevents anomalies⁸.

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First Normal Form (1NF): Each table cell should contain a single value, and each column must have a unique name⁸. No duplicate rows are present⁸.

-

Second Normal Form (2NF): All non-key attributes must depend entirely on the primary key⁸.

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Third Normal Form (3NF): Every non-key attribute must be independent of the other non-key attributes⁸. No information in the table should rely on another piece of data if it's not the primary key⁸.

-

Boyce-Codd Normal Form (BCNF): Every determinant (an attribute that can determine other attributes) must be a candidate key⁸. All dependencies are based solely on unique identifiers⁸.

-

UNION and UNION ALL operator: Used to combine results from two or more `SELECT` statements⁸....

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UNION: Combines results and removes duplicate rows⁹.

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UNION ALL: Combines results and keeps all rows, including duplicates⁹.

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Finding the second highest salary: This can be done using nested subqueries⁹. The innermost query finds the highest salary⁹. The second subquery finds the maximum salary among those less than the highest salary (thus identifying the second highest)⁹.... The outer query then selects the employees whose salary matches this second highest value¹⁰.

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Views: A **view** is like a virtual table that doesn't store data on disk but allows you to see data from one or more tables in a specific way¹⁰. Views can be used to limit sensitive data and restrict user permissions, which is helpful with large tables or when different users need access to different data parts¹⁰. Views are created using the `CREATE VIEW` statement¹⁰.

-

Converting a text into date format: The `STRING_TO_DATE` function can be used to convert a text string into a date format¹⁰. You specify the input string and the format using specifiers like `%d` for day, `%m` for month, and `%Y` for year¹⁰.

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Triggers: Triggers are like reflexive actions that allow you to set up automatic actions that run every time a certain event happens in your database, such as inserting, updating, or deleting data in a table¹¹. Triggers are usually set up to run during `INSERT`, `UPDATE`, or `DELETE` events¹¹. For example, a trigger can automatically update stock counts in a book table every time a book is borrowed or returned¹¹.... Triggers are created using the `CREATE TRIGGER` statement and specify the event (`AFTER INSERT`, `BEFORE UPDATE`, etc.), the table it acts on, and the actions to perform

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Based on the provided YouTube transcript excerpts, here is a summary of the key concepts and questions discussed regarding SQL for data analyst interviews:

The video emphasizes that **SQL is a crucial skill for various roles**, including data analysts, Power BI developers, Python developers, testers, web developers, software developers, data engineers, and data scientists¹. It highlights that **understanding SQL is often the starting point of many technical interviews** and that presenting your answers well can make a strong impression¹. The discussion is categorized into basics, next-level questions, advanced concepts, and coding questions¹.

Here are some of the key SQL concepts and interview questions covered:

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DDL (Data Definition Language) vs. DML (Data Manipulation Language):

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DDL commands change the structure of a table, such as `CREATE`, `ALTER`, and `DROP`¹. DDL commands **cannot be rolled back**¹....

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DML commands work with the actual data within a table, such as `INSERT`, `UPDATE`, and `DELETE`¹. DML commands **can be rolled back**².

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DELETE vs. DROP vs. TRUNCATE:

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TRUNCATE is a DDL command that quickly **cleans the complete table** of all records, leaving the structure intact². It **cannot be rolled back**².

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DELETE is a DML command that removes specific rows based on a `WHERE` condition or all rows if no condition is given². It **can be rolled back**².

-

DROP is a DDL command that **removes the entire table structure along with all data**². It **cannot be rolled back**².

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GROUP BY vs. ORDER BY:

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GROUP BY is used to aggregate data into buckets based on columns, typically used with aggregate functions³.

-

ORDER BY is used to **sort rows** in ascending or descending order³.

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WHERE vs. HAVING Clause:

-

The **WHERE clause** is used to **filter individual rows** based on conditions on existing columns **before grouping**³.

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The **HAVING clause** is used to **filter groups** created by `GROUP BY` based on conditions applied to aggregate results **after grouping**³.

-

Aggregate Functions: These functions perform calculations on groups of rows³. Examples include `MIN`, `MAX`, `AVERAGE`, `SUM`, and `COUNT`³. Scalar functions, like `LENGTH` or `EXTRACT`, work on a single row³....

-

COALESCE function: Used to **replace NULL values** in a column with a specified default value (e.g., 0)⁴. This is useful for calculations like `AVERAGE` that cannot handle NULLs⁴.

-

Constraints: These are rules applied to tables and columns to maintain data accuracy and reliability⁴. Types discussed include:

-

PRIMARY KEY: Uniquely identifies each row in a table⁴. It cannot have duplicate values and **cannot be NULL**⁵. There can be **only one Primary Key** per table, though it can be a combination of columns (composite key)⁵.

-

FOREIGN KEY: References the primary key in another table to link tables⁴. It can accept duplicate values and can accept NULL values⁵. A table **can have multiple Foreign Keys**⁵.

◦

UNIQUE: Ensures all values in a column are different, similar to a Primary Key, but **it can accept NULL values**⁴.... A table **can have multiple Unique constraints**⁵.

◦

DEFAULT: Assigns a default value to a column if no value is specified during insertion⁴.

◦

CHECK: Ensures that all values in a column satisfy a specific condition⁴.

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SQL Joins: Used to combine rows from two or more tables based on a related column⁵. Types mentioned include:

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INNER JOIN: Returns **only the matching records** from both tables⁵....

◦

LEFT JOIN: Returns **all records from the left table**, and the matching records from the right table; non-matching records from the right table will have NULL values⁶.

◦

RIGHT JOIN: Returns **all records from the right table**, and the matching records from the left table; non-matching records from the left table will have NULL values⁶.

◦

FULL JOIN: Returns **all records** from both tables, including matching and non-matching ones, with NULLs where there is no match⁶. It is often avoided in applications due to potential large result sets⁶.

◦

SELF JOIN: Used to **join a table with itself**⁶.

◦

CROSS JOIN (Cartesian Join): Combines every row of the first table with every row of the second table⁶.... This significantly increases the result set size⁶. An example given is combining different t-shirt sizes and colors⁷....

◦

To join 'n' tables, you generally need **n-1 joins**⁹.

•

UNION vs. UNION ALL: Used to combine result sets of two or more `SELECT` statements¹⁰.

◦

UNION: Combines results and **removes duplicate rows**¹⁰. It is slower than `UNION ALL` due to the duplicate removal process¹⁰.

◦

UNION ALL: Combines results and **includes all rows, including duplicates**¹⁰. It is faster than `UNION`¹⁰. Both require the tables to have an equal number of columns with matching data types¹⁰.

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Order of Execution: The sequence in which SQL clauses are processed¹⁰. The typical order is: **FROM**, then **WHERE**, then **GROUP BY** and **HAVING**, then **SELECT**, and finally **ORDER BY** and **LIMIT**¹⁰.

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Window Functions (Analytical Functions): Functions that perform calculations across a set of table rows that are somehow related to the current row⁹. Examples include `ROW_NUMBER`, `RANK`, `DENSE_RANK`, `LEAD`, and `LAG`⁹.

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RANK vs. DENSE_RANK: When ranking values with ties, `RANK` skips the next rank number, while `DENSE_RANK` does not⁹. For example, scores 99, 98, 98, 97 would get ranks 1, 2, 2, 4 with `RANK` and 1, 2, 2, 3 with `DENSE_RANK`⁹.

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View: A **virtual table** that does not store data itself but stores a query¹¹. When a view is queried, the stored query is executed to display results¹¹. Views can be used to simplify complex queries or restrict access to certain data¹¹.

-

Indexing: A database object created on a column to **speed up data retrieval**¹¹. An index helps the database quickly locate relevant data without scanning the entire table (full scan)¹¹. A **Clustered Index** physically orders the data rows in the table according to the index¹¹.

-

CTE (Common Table Expression): A temporary, named result set that can be referenced within a single execution statement⁷. CTEs start with the `WITH` clause and help in breaking down complex queries into simpler, readable parts without creating temporary tables⁷.

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SQL vs. PL/SQL:

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SQL consists of basic commands for data manipulation and definition, executed line by line⁷.

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PL/SQL (Procedural Language SQL) adds procedural programming concepts like `FOR` loops, `WHILE` loops, and `IF` conditions, allowing code to be executed in blocks⁷. Procedures and functions are part of PL/SQL⁷.

The video also briefly mentions coding questions often asked in interviews and approaches to solve them⁸:

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Finding the **Nth highest salary** (e.g., third maximum salary) can be achieved using window functions like `RANK` or `DENSE_RANK` within a subquery or CTE⁸.

-

Finding and removing **duplicate records** can be done by using `GROUP BY` with `COUNT(*)` and `HAVING COUNT(*) > 1` to identify duplicates, and then using `DELETE` with an `IN` condition to remove them⁸.

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Finding **Top N records** (e.g., top 5 states by sales) can be done using `ORDER BY` and `LIMIT` or by using `RANK/DENSE_RANK` window functions⁸.

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Displaying **alternative rows** (odd or even rows) can be done using the modulus operator (%) on an ID column⁸.

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Working with **date columns** involves using functions like `EXTRACT` to filter data by year, month, etc., or using date arithmetic (e.g., subtracting days for recent periods)⁸....

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Handling **hierarchical data** like manager-employee relationships in a single table can be solved using a **Self Join**¹².